

SHOPFORGE

Ecommerce Platform

Database Design Document (DDD)

PostgreSQL 16 | Version 1.0

1. Design Conventions

Convention	Rule
Primary Keys	UUID v4 (gen_random_uuid()) on all tables – no integer sequences exposed to API
Timestamps	created_at TIMESTAMPTZ DEFAULT now(), updated_at TIMESTAMPTZ – all UTC
Soft Delete	deleted_at TIMESTAMPTZ NULL – NULL means active, non-null means deleted
Naming	snake_case tables and columns; table names plural
Enum Types	PostgreSQL ENUM types for finite state sets
Money	NUMERIC(12,2) for all monetary values – never FLOAT
Text	VARCHAR with explicit limits where bounded; TEXT for unbounded
Indexes	B-tree default; GIN for arrays/JSONB/FTS; BRIN for append-

Convention	Rule
	only timestamp cols
Constraints	FK constraints enforced at DB level; CHECK constraints on enums and ranges
Schema	All tables in public schema; separate schema 'audit' for audit logs

2. Entity Relationship Overview

Core entity groups and their relationships:

Group	Tables	Notes
Identity	users, user_roles, oauth_providers, refresh_tokens	Auth and identity
Catalog	categories, products, product_variants, product_images, product_tags, tags, brands	Product data
Inventory	inventory, inventory_transactions	Stock tracking
Commerce	carts, cart_items, wishlists, wishlist_items	Pre-purchase
Orders	orders, order_items, order_status_history, shipments, returns, return_items	Post-purchase
Payments	payments, payment_methods, refunds	Financial
Addresses	addresses	Shipping/billing
Reviews	reviews, review_images, review_votes	UGC
Promotions	coupons, coupon_usages, flash_sales, flash_sale_products	Marketing
Notifications	notifications, notification_templates	Comms
Audit	audit.audit_logs	Compliance

3. Table Schemas

3.1 users

Core identity table. Supports email+password and OAuth login.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
email	VARCHAR(255)	NO	—	Unique email, lowercase-normalized
email_verified	BOOLEAN	NO	false	True after OTP verification
phone	VARCHAR(20)	YES	NULL	E.164 format: +919876543210
phone_verified	BOOLEAN	NO	false	True after SMS OTP
password_hash	VARCHAR(255)	YES	NULL	bcrypt hash; NULL for OAuth-only
first_name	VARCHAR(100)	NO	—	Given name
last_name	VARCHAR(100)	YES	NULL	Family name
avatar_url	TEXT	YES	NULL	S3 URL of profile image
status	user_status	NO	ACTIVE	ENUM: ACTIVE, BANNED, DELETED
failed_login_attempts	SMALLINT	NO	0	Reset on successful login
lockout_until	TIMESTAMPTZ	YES	NULL	NULL if not locked out
last_login_at	TIMESTAMPTZ	YES	NULL	Updated on each login
created_at	TIMESTAMPTZ	NO	now()	Account creation time
updated_at	TIMESTAMPTZ	NO	now()	Last profile update
deleted_at	TIMESTAMPTZ	YES	NULL	Soft delete timestamp

Indexes:

- UNIQUE on email (case-insensitive via functional index on lower(email))
- UNIQUE on phone WHERE phone IS NOT NULL
- INDEX on status WHERE deleted_at IS NULL

3.2 user_roles

Column	Type	Nullable	Default	Description
user_id	UUID	NO	—	FK → users.id ON DELETE CASCADE
role	user_role	NO	—	ENUM: CUSTOMER, SELLER, ADMIN, SUPPORT
granted_at	TIMESTAMPTZ	NO	now()	When role assigned
granted_by	UUID	YES	NULL	FK → users.id (admin who granted)

PK: (user_id, role)

3.3 oauth_providers

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
user_id	UUID	NO	—	FK → users.id ON DELETE CASCADE
provider	VARCHAR(50)	NO	—	google, facebook, apple
provider_user_id	VARCHAR(255)	NO	—	ID from provider
access_token	TEXT	YES	NULL	Encrypted; may be NULL
refresh_token	TEXT	YES	NULL	Encrypted
expires_at	TIMESTAMPTZ	YES	NULL	Token expiry
created_at	TIMESTAMPTZ	NO	now()	—

UNIQUE on (provider, provider_user_id)

3.4 refresh_tokens

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
user_id	UUID	NO	—	FK → users.id ON DELETE CASCADE
token_hash	VARCHAR(255)	NO	—	SHA-256 of actual token
expires_at	TIMESTAMPTZ	NO	—	Token expiry datetime
revoked	BOOLEAN	NO	false	True if explicitly revoked
device_info	VARCHAR(500)	YES	NULL	User-agent string
ip_address	INET	YES	NULL	Client IP at issue time
created_at	TIMESTAMPTZ	NO	now()	—

3.5 categories

Self-referential tree (adjacency list). Max 3 levels recommended.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
parent_id	UUID	YES	NULL	FK → categories.id; NULL = root
name	VARCHAR(200)	NO	—	Display name
slug	VARCHAR(220)	NO	—	URL-safe unique identifier
description	TEXT	YES	NULL	Optional description
image_url	TEXT	YES	NULL	Category banner image

Column	Type	Nullable	Default	Description
sort_order	INT	NO	0	Display order within parent
is_active	BOOLEAN	NO	true	False = hidden from storefront
meta_title	VARCHAR(255)	YES	NULL	SEO title override
meta_description	TEXT	YES	NULL	SEO meta description
created_at	TIMESTAMPTZ	NO	now()	—
updated_at	TIMESTAMPTZ	NO	now()	—

UNIQUE on slug. INDEX on parent_id.

3.6 brands

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
name	VARCHAR(200)	NO	—	Brand name
slug	VARCHAR(220)	NO	—	UNIQUE URL slug
logo_url	TEXT	YES	NULL	S3 logo URL
description	TEXT	YES	NULL	About the brand
is_active	BOOLEAN	NO	true	—
created_at	TIMESTAMPTZ	NO	now()	—

3.7 products

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
seller_id	UUID	NO	—	FK → users.id (must have SELLER role)
category_id	UUID	NO	—	FK → categories.id
brand_id	UUID	YES	NULL	FK → brands.id
name	VARCHAR(500)	NO	—	Product title
slug	VARCHAR(550)	NO	—	UNIQUE URL slug
description	TEXT	YES	NULL	Rich text / HTML
short_description	VARCHAR(500)	YES	NULL	Summary text
status	product_status	NO	DRAFT	ENUM: DRAFT, ACTIVE, INACTIVE, ARCHIVED
is_featured	BOOLEAN	NO	false	Show on homepage
has_variants	BOOLEAN	NO	false	True if SKU-level variants exist

Column	Type	Nullable	Default	Description
base_price	NUMERIC(12,2)	NO	—	Default price (overridden by variant)
sale_price	NUMERIC(12,2)	YES	NULL	Sale price; NULL = no sale
tax_class	VARCHAR(50)	YES	NULL	Tax profile code
weight_grams	INT	YES	NULL	For shipping calc
meta_title	VARCHAR(255)	YES	NULL	SEO
meta_description	TEXT	YES	NULL	SEO
search_vector	TSVECTOR	YES	NULL	FTS vector; updated via trigger
view_count	BIGINT	NO	0	Denormalized for performance
avg_rating	NUMERIC(3,2)	YES	NULL	Denormalized; updated on review
review_count	INT	NO	0	Denormalized count
created_at	TIMESTAMPTZ	NO	now()	—
updated_at	TIMESTAMPTZ	NO	now()	—
deleted_at	TIMESTAMPTZ	YES	NULL	Soft delete

Indexes:

- UNIQUE on slug
- INDEX on (category_id, status) WHERE deleted_at IS NULL
- INDEX on seller_id WHERE deleted_at IS NULL
- GIN INDEX on search_vector
- INDEX on avg_rating DESC, review_count DESC (for sort by rating)

3.8 product_variants

Each variant is a purchasable SKU with its own price and stock.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
product_id	UUID	NO	—	FK → products.id ON DELETE CASCADE
sku	VARCHAR(100)	NO	—	UNIQUE stock keeping unit
name	VARCHAR(300)	YES	NULL	e.g. Red / XL; auto-generated if NULL
attributes	JSONB	YES	'{}'	e.g. {"color":"red","size":"XL"}
price	NUMERIC(12,2)	NO	—	Variant-specific price
sale_price	NUMERIC(12,2)	YES	NULL	Variant sale price
cost_price	NUMERIC(12,2)	YES	NULL	Internal cost (not shown to customers)
barcode	VARCHAR(100)	YES	NULL	EAN/UPC barcode
weight_grams	INT	YES	NULL	Override product weight
is_active	BOOLEAN	NO	true	Can be purchased
sort_order	INT	NO	0	Display order

Column	Type	Nullable	Default	Description
created_at	TIMESTAMPTZ	NO	now()	—
updated_at	TIMESTAMPTZ	NO	now()	—

UNIQUE on sku. GIN INDEX on attributes.

3.9 product_images

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
product_id	UUID	NO	—	FK → products.id ON DELETE CASCADE
variant_id	UUID	YES	NULL	FK → product_variants.id; NULL = applies to all
url	TEXT	NO	—	S3 CDN URL (WebP)
alt_text	VARCHAR(255)	YES	NULL	Accessibility alt text
is_primary	BOOLEAN	NO	false	Main listing image
sort_order	INT	NO	0	Gallery order
created_at	TIMESTAMPTZ	NO	now()	—

CHECK: only one is_primary=true per product (enforced in app + partial unique index).

3.10 inventory

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
variant_id	UUID	NO	—	FK → product_variants.id UNIQUE
quantity_on_hand	INT	NO	0	Current physical stock
quantity_reserved	INT	NO	0	Held for pending orders
quantity_available	INT	NO	—	GENERATED: on_hand - reserved
reorder_threshold	INT	NO	10	Trigger low-stock alert below this
reorder_quantity	INT	YES	NULL	Suggested reorder amount
updated_at	TIMESTAMPTZ	NO	now()	—

quantity_available is a GENERATED ALWAYS AS (quantity_on_hand - quantity_reserved) STORED column.

UNIQUE on variant_id. CHECK quantity_on_hand >= 0 AND quantity_reserved >= 0.

3.11 inventory_transactions

Append-only ledger. Never update, only insert.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
variant_id	UUID	NO	—	FK → product_variants.id
type	inv_txn_type	NO	—	ENUM: PURCHASE, SALE, RETURN, ADJUSTMENT, RESERVATION, RELEASE
quantity_delta	INT	NO	—	Positive = stock in, negative = stock out
reference_type	VARCHAR(50)	YES	NULL	order, return, adjustment
reference_id	UUID	YES	NULL	FK to referenced entity
note	TEXT	YES	NULL	Manual adjustment note
created_by	UUID	YES	NULL	FK → users.id
created_at	TIMESTAMPTZ	NO	now()	—

BRIN INDEX on created_at. INDEX on variant_id.

3.12 carts

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
user_id	UUID	YES	NULL	FK → users.id; NULL = guest cart
session_id	VARCHAR(255)	YES	NULL	Guest session identifier
coupon_id	UUID	YES	NULL	FK → coupons.id (applied coupon)
expires_at	TIMESTAMPTZ	NO	—	Auto-expire inactive carts
created_at	TIMESTAMPTZ	NO	now()	—
updated_at	TIMESTAMPTZ	NO	now()	—

3.13 cart_items

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
cart_id	UUID	NO	—	FK → carts.id ON DELETE CASCADE
variant_id	UUID	NO	—	FK → product_variants.id
quantity	INT	NO	1	CHECK quantity > 0
unit_price	NUMERIC(12,2)	NO	—	Price at time of add (snapshot)

Column	Type	Nullable	Default	Description
created_at	TIMESTAMPTZ	NO	now()	—
updated_at	TIMESTAMPTZ	NO	now()	—

UNIQUE on (cart_id, variant_id).

3.14 addresses

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
user_id	UUID	NO	—	FK → users.id ON DELETE CASCADE
label	VARCHAR(100)	YES	NULL	Home, Work, etc.
first_name	VARCHAR(100)	NO	—	Recipient first name
last_name	VARCHAR(100)	YES	NULL	Recipient last name
phone	VARCHAR(20)	NO	—	Contact phone for delivery
address_line1	VARCHAR(300)	NO	—	Street address
address_line2	VARCHAR(300)	YES	NULL	Apt, Suite, etc.
city	VARCHAR(100)	NO	—	City
state	VARCHAR(100)	NO	—	State / Province
country	VARCHAR(2)	NO	IN	ISO 3166-1 alpha-2
postal_code	VARCHAR(20)	NO	—	PIN/ZIP code
is_default	BOOLEAN	NO	false	Default shipping address
created_at	TIMESTAMPTZ	NO	now()	—

3.15 orders

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
order_number	VARCHAR(30)	NO	—	UNIQUE human-readable: ORD-20240115-XXXX
user_id	UUID	YES	NULL	FK → users.id; NULL = guest order
guest_email	VARCHAR(255)	YES	NULL	For guest orders
status	order_status	NO	PENDING	See status flow in SRS
shipping_address_id	UUID	YES	NULL	FK → addresses.id
shipping_address_snapshot	JSONB	NO	—	Address at time of order (immutable)

Column	Type	Nullable	Default	Description
shipping_method	VARCHAR(100)	YES	NULL	Standard, Express, etc.
shipping_cost	NUMERIC(12,2)	NO	0.00	Shipping fee charged
subtotal	NUMERIC(12,2)	NO	—	Sum of item prices
discount_amount	NUMERIC(12,2)	NO	0.00	Coupon/promo reduction
tax_amount	NUMERIC(12,2)	NO	0.00	Tax applied
total_amount	NUMERIC(12,2)	NO	—	Final charged amount
coupon_id	UUID	YES	NULL	FK → coupons.id
coupon_code	VARCHAR(50)	YES	NULL	Snapshot of code used
notes	TEXT	YES	NULL	Customer delivery notes
ip_address	INET	YES	NULL	Order placed from IP
created_at	TIMESTAMPTZ	NO	now()	—
updated_at	TIMESTAMPTZ	NO	now()	—

UNIQUE on order_number. INDEX on user_id. INDEX on status. INDEX on created_at DESC.

3.16 order_items

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
order_id	UUID	NO	—	FK → orders.id ON DELETE RESTRICT
variant_id	UUID	NO	—	FK → product_variants.id
product_snapshot	JSONB	NO	—	Product/variant data at purchase time
quantity	INT	NO	—	CHECK > 0
unit_price	NUMERIC(12,2)	NO	—	Price per unit at purchase
total_price	NUMERIC(12,2)	NO	—	unit_price * quantity
tax_amount	NUMERIC(12,2)	NO	0.00	Tax on this line item
seller_id	UUID	NO	—	FK → users.id (snapshot of seller)
fulfillment_status	VARCHAR(50)	NO	PENDING	Per-item fulfillment

3.17 order_status_history

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
order_id	UUID	NO	—	FK → orders.id ON DELETE CASCADE

Column	Type	Nullable	Default	Description
from_status	order_status	YES	NULL	Previous status
to_status	order_status	NO	—	New status
changed_by	UUID	YES	NULL	FK → users.id
note	TEXT	YES	NULL	Optional admin note
created_at	TIMESTAMPTZ	NO	now()	—

3.18 shipments

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
order_id	UUID	NO	—	FK → orders.id
tracking_number	VARCHAR(200)	YES	NULL	Courier tracking code
carrier	VARCHAR(100)	YES	NULL	Delhivery, BlueDart, etc.
carrier_url	TEXT	YES	NULL	Tracking URL template
shipped_at	TIMESTAMPTZ	YES	NULL	Dispatch timestamp
expected_delivery	DATE	YES	NULL	Estimated delivery date
delivered_at	TIMESTAMPTZ	YES	NULL	Actual delivery timestamp
created_at	TIMESTAMPTZ	NO	now()	—

3.19 payments

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
order_id	UUID	NO	—	FK → orders.id UNIQUE (one payment per order)
gateway	VARCHAR(50)	NO	—	razorpay, stripe, cod
gateway_order_id	VARCHAR(255)	YES	NULL	Gateway's order/session ID
gateway_payment_id	VARCHAR(255)	YES	NULL	Gateway's payment ID
gateway_signature	VARCHAR(500)	YES	NULL	Webhook signature for verification
method	VARCHAR(50)	YES	NULL	upi, card, netbanking, wallet, cod
amount	NUMERIC(12,2)	NO	—	Amount charged
currency	VARCHAR(3)	NO	INR	ISO 4217 currency code
status	payment_status	NO	INITIATED	ENUM: INITIATED, SUCCESS, FAILED, REFUNDED, PARTIALLY_REFUNDED

Column	Type	Nullable	Default	Description
idempotency_key	VARCHAR(255)	NO	—	UNIQUE — prevent double-charge
paid_at	TIMESTAMPTZ	YES	NULL	Confirmation timestamp
created_at	TIMESTAMPTZ	NO	now()	—
updated_at	TIMESTAMPTZ	NO	now()	—

3.20 refunds

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
payment_id	UUID	NO	—	FK → payments.id
amount	NUMERIC(12,2)	NO	—	Refund amount
reason	TEXT	YES	NULL	Reason for refund
gateway_refund_id	VARCHAR(255)	YES	NULL	Gateway refund reference
status	VARCHAR(50)	NO	PENDING	PENDING, SUCCESS, FAILED
processed_by	UUID	YES	NULL	FK → users.id (admin)
created_at	TIMESTAMPTZ	NO	now()	—
updated_at	TIMESTAMPTZ	NO	now()	—

3.21 reviews

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
product_id	UUID	NO	—	FK → products.id
user_id	UUID	NO	—	FK → users.id
order_item_id	UUID	YES	NULL	FK → order_items.id (verified purchase)
rating	SMALLINT	NO	—	CHECK rating BETWEEN 1 AND 5
title	VARCHAR(200)	YES	NULL	Review headline
body	TEXT	NO	—	CHECK length(body) >= 20
status	review_status	NO	PENDING	ENUM: PENDING, APPROVED, REJECTED, FLAGGED
is_verified_purchase	BOOLEAN	NO	false	True if order_item_id set
helpful_count	INT	NO	0	Denormalized vote count
created_at	TIMESTAMPTZ	NO	now()	—
updated_at	TIMESTAMPTZ	NO	now()	—

UNIQUE on (product_id, user_id) — one review per product per user.

INDEX on (product_id, status). INDEX on rating.

3.22 coupons

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
code	VARCHAR(50)	NO	—	UNIQUE; uppercase
type	coupon_type	NO	—	ENUM: PERCENTAGE, FIXED_AMOUNT, FREE_SHIPPING, BUY_X_GET_Y
value	NUMERIC(12,2)	NO	—	Discount value (% or amount)
min_order_value	NUMERIC(12,2)	YES	NULL	Minimum cart value to apply
max_discount_amount	NUMERIC(12,2)	YES	NULL	Cap on PERCENTAGE discounts
usage_limit	INT	YES	NULL	Total uses allowed; NULL = unlimited
usage_count	INT	NO	0	Current use count
per_user_limit	INT	YES	NULL	Uses per user; NULL = unlimited
applicable_to	VARCHAR(50)	NO	ALL	ALL, CATEGORY, PRODUCT
applicable_ids	UUID[]	YES	NULL	Category or product IDs if scoped
starts_at	TIMESTAMPTZ	NO	—	Coupon activation time
expires_at	TIMESTAMPTZ	YES	NULL	Expiry; NULL = no expiry
is_active	BOOLEAN	NO	true	—
created_by	UUID	NO	—	FK → users.id (admin)
created_at	TIMESTAMPTZ	NO	now()	—

INDEX on (code, is_active). GIN INDEX on applicable_ids.

3.23 notifications

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	PK
user_id	UUID	NO	—	FK → users.id ON DELETE CASCADE
type	VARCHAR(100)	NO	—	ORDER_SHIPPED, PAYMENT_CONFIRMED, etc.
title	VARCHAR(300)	NO	—	Notification title
body	TEXT	YES	NULL	Notification body
data	JSONB	YES	'{}'	Extra payload for deep linking
read_at	TIMESTAMPTZ	YES	NULL	NULL = unread

Column	Type	Nullab le	Default	Description
created_at	TIMESTAMPTZ	NO	now()	—

INDEX on (user_id, read_at) WHERE read_at IS NULL — fast unread count.

4. PostgreSQL ENUM Types

All ENUM types created before table creation:

```
CREATE TYPE user_status AS ENUM ('ACTIVE', 'BANNED', 'DELETED');
CREATE TYPE user_role AS ENUM ('CUSTOMER', 'SELLER', 'ADMIN', 'SUPPORT');
CREATE TYPE product_status AS ENUM ('DRAFT', 'ACTIVE', 'INACTIVE', 'ARCHIVED');
CREATE TYPE order_status AS ENUM ('PENDING', 'PAYMENT_RECEIVED', 'PROCESSING',
'SHIPPED', 'DELIVERED', 'COMPLETED', 'CANCELLED', 'RETURN_REQUESTED',
'RETURN_APPROVED', 'RETURN_REJECTED', 'REFUND_INITIATED', 'REFUNDED');
CREATE TYPE payment_status AS ENUM ('INITIATED', 'SUCCESS', 'FAILED', 'REFUNDED',
'PARTIALLY_REFUNDED');
CREATE TYPE coupon_type AS ENUM ('PERCENTAGE', 'FIXED_AMOUNT', 'FREE_SHIPPING',
'BUY_X_GET_Y');
CREATE TYPE review_status AS ENUM ('PENDING', 'APPROVED', 'REJECTED', 'FLAGGED');
CREATE TYPE inv_txn_type AS ENUM ('PURCHASE', 'SALE', 'RETURN', 'ADJUSTMENT',
'RESERVATION', 'RELEASE');
```

5. Key Database Triggers & Functions

Trigger	Table	Event	Purpose
update_search_vector	products	INSERT / UPDATE	Recompute tsvector from name + description + brand
update_product_rating	reviews	INSERT / UPDATE / DELETE	Recalculate avg_rating and review_count on products
set_updated_at	All tables with updated_at	UPDATE	Auto-set updated_at = now()
inventory_consistency	inventory_transactions	INSERT	Apply delta to inventory.quantity_on_hand
order_number_gen	orders	INSERT BEFORE	Generate ORD-YYYYMMDD-{random4} order number

Search vector trigger example:

```
CREATE FUNCTION update_search_vector() RETURNS TRIGGER AS $$
BEGIN
    NEW.search_vector := to_tsvector('english',
        coalesce(NEW.name, '') || ' ' ||
        coalesce(NEW.short_description, '') ||
        coalesce(NEW.description, ''));
    RETURN NEW;
END; $$ LANGUAGE plpgsql;
```

6. Index Strategy Summary

Table	Index Type	Columns	Purpose
users	UNIQUE B-tree	lower(email)	Case-insensitive email lookup
products	GIN	search_vector	Full-text search
products	B-tree	(category_id, status)	Category listing filter
products	B-tree	slug	URL routing
product_variants	GIN	attributes	Attribute filter queries
inventory	B-tree	variant_id (UNIQUE)	Stock lookup by SKU
orders	B-tree	(user_id, created_at DESC)	User order history
orders	B-tree	status	Admin order queue filters
order_items	B-tree	order_id	Order detail join
payments	UNIQUE B-tree	idempotency_key	Prevent duplicate payments
reviews	UNIQUE B-tree	(product_id, user_id)	One review per user/product
notifications	B-tree	(user_id) WHERE read_at IS NULL	Unread count query
inventory_transactions	BRIN	created_at	Time-range scans on large table
coupons	B-tree	(code, is_active)	Coupon validation lookup
refresh_tokens	B-tree	token_hash	Token lookup on refresh

7. Migration Strategy

- Flyway for schema versioning (V1__init.sql, V2__add_brands.sql etc.)
- All migrations additive in Phase 1 — no DROP COLUMN without deprecation period
- Test migrations on staging DB before prod — use Flyway validate in CI
- Data migrations in separate numbered scripts (R__ prefix for repeatable)
- Zero-downtime: add nullable columns, backfill, add NOT NULL constraint in next deploy

8. Backup & Recovery

Strategy	Detail
Automated Snapshots	Daily pg_dump via pg_cron to S3; 30-day retention
WAL Archiving	PITR enabled; recover to any point in last 7 days
Read Replica	Streaming replica for reporting queries; zero write load on primary
RT0 Target	< 4 hours for full restore; < 30 min for PITR
RPO Target	< 5 minutes data loss
Test Restores	Monthly restore drill to staging environment